

Project Presentation Outline: CS 466

There is no time duration or limit. So, there is no need to rush through the slides.

Project Presentation Guidelines (CSC 466)

The following guidelines describe how to structure your PowerPoint presentation. These are general suggestions and may be adapted depending on the specific problem you are investigating. However, you should aim to organize your presentation using the following sections.

Example:

1. Introduction / Motivation
2. Problem Definition
3. Dataset
4. Related Work
5. Algorithms / Methods
6. Experimental Setup and Results
7. New Analysis / Insights
8. Conclusion

1. Introduction

Motivate and describe the problem you are addressing at a high level. Explain **why the problem is interesting or important** and briefly outline your overall approach.

Clearly highlight **what you have done differently** from previous work or earlier stages of the project. In particular:

- Identify any **new KDD step** you introduced (e.g., new preprocessing, feature engineering, model comparison, evaluation technique).
- Explain how your approach **differs from the earlier implementation**.
- Demonstrate **new analysis**, such as:
 - additional visualizations,
 - deeper exploratory data analysis,
 - a new model comparison,
 - or a unique insight derived from the data.

Your presentation should clearly demonstrate at least one **new analysis step** (e.g., additional visualization, feature engineering, model comparison, or deeper data interpretation).

2. Problem Definition and Algorithms

Task Definition

Clearly define the problem you are addressing.

- Specify the **input data and expected outputs**.
- Define any **notation** you will use.
- Explain why this task is meaningful or relevant.

Algorithm(s) Used

Describe the algorithm(s) used in your analysis.

- Explain the intuition behind each algorithm.
- Provide sufficient detail so that the audience understands **how the method works**.
- Pseudocode or simple workflow diagrams may be useful.
- If helpful, walk through a **small example** illustrating how the algorithm operates.

Your Contribution / Expectations

Discuss what you expected to observe before running the experiments.

- What was your **hypothesis**?
- How did you expect the algorithms to behave?
- Why did you expect those outcomes?

Explain how these expectations influenced your **experimental design**.

3. Dataset

Include:

- dataset description
- training / validation / test split
- preprocessing steps
- evaluation metrics (accuracy, F1, precision, recall, etc.)

Explain why the chosen methodology is **appropriate for the problem**.

4. Related Work if any

Briefly discuss relevant research or prior work.

- What approaches have others used to solve this problem?
- How does your method compare?
- What distinguishes your approach from existing methods?

This section should demonstrate **awareness of existing approaches**, but it does not need to be exhaustive.

5. Algorithm Theoretical Background

If your work involves mathematical models or theoretical concepts:

- briefly introduce the **key equations or theory**
- explain **what the equations mean intuitively**
- avoid presenting equations without interpretation

The goal is to help the audience **understand the idea**, not just see the formula.

6. Methodology: Results and Experimental Evaluation

Describe how you evaluated your approach.

Present the **quantitative results** of your experiments.

- Compare results across algorithms or models.
- When possible, compare against **baseline methods**.
- Use **visualizations (graphs, charts, confusion matrices)** to present results clearly.
- Graphical presentations are often more informative than large tables.

Discussion

Interpret the results.

Discuss:

- whether your **hypothesis was supported**
- strengths and weaknesses of the methods
- possible reasons for the observed performance

Explain how the results relate to the **properties of the algorithms and the dataset**.

Data Analysis

Provide deeper analysis where appropriate.

Examples include:

- error analysis
- examining misclassified examples
- identifying patterns in model failures
- explaining surprising outcomes

This section helps demonstrate **critical thinking and deeper understanding** of the data.

7. New Analysis/Insights

Examples:

- improving model performance
- using larger or different datasets
- testing additional algorithms
- improving feature engineering

Explain any improvement or contribution that you have added.

how these improvements could address current limitations.

8. Future Work & Conclusion

If applicable, discuss possible improvements or things that you could have done

Summarize the key points of your work.

Focus on:

- the problem you addressed
- your approach
- the main results
- the most important insights gained from your analysis